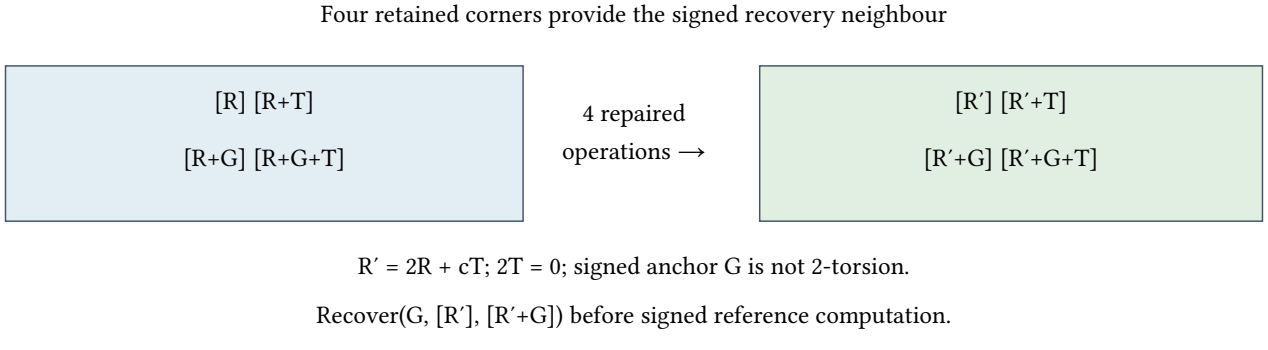

Four-corner state with propagated recovery neighbour

The private construction passes252 consecutive fixed transitions across84 fixture cases and21 models. All1008 quotient corners match signed references, and all252 recovered divisors equal the signed expected point. Recovery uses the propagated non-torsion neighbour before the test harness computes its signed reference. This closes the supplied-neighbour dependency for this tested transition family. It does not implement or benchmark the original generic G-step collector.

State and derived transition

Assume a valid Jacobian, nonzero rational T with2T=0, signed anchor G not in2-torsion, rQ=rG=0, and canonical labels moduloN=2r<=2^63. R=aT+bQ. Retain four quotient points [R],[R+T],[R+G],[R+G+T], plus fixed [0],[T],[G],[G+T]. Initialization constructs the needed signed sums and projects them; its source counters are separate.

Compute A=[2R] by doubling, B=[2R+T] from [R],[R+T] with difference[T], C=[2R+G] from [R],[R+G] with difference[G], and D=[2R+G+T] from [R],[R+G+T] with difference[G+T]. Each call uses repaired differential arithmetic. With c=branch([R]), c0 returns[A,B,C,D]; c1 returns[B,A,D,C]. The second permutation uses2T=0. Therefore the returned third entry is precisely[R'+G] for R'=2R+cT. Labels update to(2a+c,2b) moduloN; canonical labels and N<=2^63 bound both word updates, and torsion hypotheses justify modular reduction.

The helper grid_step accepts only quotient arrays, fixed quotient differences, field/model constants and repair actions. No signed expected point enters it. Signed recovery receives G and returned entries0/2. The reference is computed afterward solely for verification. Source structure and phase counters retain this order.

Failure preserved and bounded Delta correction

The first four-corner run already matched all1008 corners and recovered251 of252 points. At p739, exception20, step1, generic recovery returned Delta_zero for expected degree2 u=[512,518,1],v=[263,393]. That all-pass gate exited101 and remains immutable in the original four-corner packet.

The additive fallback runs only on Delta_zero. For $u=X^2+a1X+a0$, derive $B=b1^2$ from the existing classical fourth-coordinate equation. Compute f modulo u=(r0,r1). For each square root b1 of B, if b1!=0 use b0=(r1+a1B)/(2b1); if b1=0 use square roots of r0. Deduplicate at most four candidates. Require valid Mumford representation, project-back equality, and unique match after signed-anchor addition to the propagated neighbour. No field, group or scalar enumeration occurs. Existing D_zero fallback is unchanged.

The additive run recovers243 points through the generic path,8 through the degree1 chart and1 through the new Delta fallback. The Delta case tests2 candidates and returns exactly the expected divisor. Fixed zero-input, zero-neighbour and inconsistent-neighbour controls reject. Ambiguous and nonsquare Delta branches are explicit but unexercised in this packet.

Accounting and interpretation

Every transition records all four repaired calls, not the cheaper two-call proposal. Initialization, transition, recovery and reference verification have separate native field/walk/signed counter records. phase-counters.csv has252 rows; initialization-counters.csv has84. Transition walk counters vary with repair attempts: (M,A,I)=(169,228,21) for48 rows, (225,296,25) for78,(281,364,29) for6,and(337,432,33) for120. These include branch selection and four repaired operations. They are native source-counter observations, not elapsed timing or a complete operation census; exponentiation internals are not uniformly tallied across counter banks. Never sum overlapping counters as independent phases without

inspecting definitions. Model construction, subgroup qualification and standalone negative-control work are outside initialization/transition/recovery columns.

No scientific solve/cohort or matched-rho comparison was run. Four retained corners and four differential calls are a different state/cost contract from the prior two-call proposal. The assumptions and algebraic closure are derived; this packet supplies finite source checks, not a universal proof of coordinate correctness. Independent review is pending. Remaining collector work includes target-label relation semantics, complete pricing and a prospectively reviewed experiment.

Reproducibility and source lineage

PLAN.md freezes the formulas, assumptions and checks before code. pre-execution-manifest.json pins inputs and source; receipt.json records the120s locked/offline command and exit0. Original252/251 failure path: /var/tmp/pencil-kummer-four-corner-20261031. Earlier provided-neighbour qualification is corrected by /var/tmp/pencil-kummer-recovery-scope-addendum-20261031. No original evidence, shared source, cloud job, runtime or deployment was changed.

Literature: Chung-Costello-Smith sections2,6.3,6.4,7.4, https://www.microsoft.com/en-us/research/wp-content/uploads/2016/02/recovery.pdf . Local sources: pinned N19 main_labelled.rs, walk_impl_uncached.rs, recover_impl_uncached.rs and additive degree1 implementation. No novelty or speed claim is made.